SIDDHANTH MUTHA

AI Engineer | Generative AI & Full-Stack Development

Pune, Maharashtra, India | siddhanthm12@gmail.com | +91 82083 61516 | LinkedIn | GitHub

Software engineer with 2+ years of production experience across full-stack development and Generative AI. Hands-on expertise building Retrieval-Augmented Generation (RAG) systems and multi-agent Large Language Model (LLM) architectures with LangChain and LangGraph. Published IEEE researcher in machine learning forecasting, with a track record of shipping production systems end-to-end.

EXPERIENCE

IT Engineer — Canpex Chemical Pvt Ltd *September 2024 – Present*

- Architected a custom ERP system from scratch, designing the complete MySQL database schema to replace an off-the-shelf solution for 3 business departments.
- Developed a full-stack web application with React and Node.js, delivering RESTful APIs that connected the database to the UI for 25+ daily users.
- Created interactive dashboards and data visualization components for real-time business metrics, enabling data-driven decisions across departments.
- Automated routine IT operations with Python and shell scripting, including backups, log rotation, and user provisioning, cutting manual effort by 60%.

Software Development Intern — Parkar Digital Pvt Ltd *January 2024 – August 2024*

- Developed and optimized RESTful APIs in Python to connect React frontends with backend services, ensuring secure and efficient data exchange.
- Performed data cleaning, transformation, and analysis in Python to prepare datasets for business intelligence applications.
- Wrote and tuned complex SQL queries, improving data storage, retrieval, and performance across relational databases.

PROJECTS

FuseRAG — **Production-minded RAG system**

- Engineered a Retrieval-Augmented Generation (RAG) pipeline fusing semantic vector search with BM25 keyword matching via Reciprocal Rank Fusion. Added cross-encoder reranking to improve context relevance.
- Implemented an evaluation-first framework across 6 metrics, including Precision@k, Recall@k, MRR, nDCG, Faithfulness, and Correctness. Reduced hallucination risk through automated output validation.
- Delivered a FastAPI backend integrating a Pinecone-compatible vector store and PostgreSQL/pgvector metadata storage, with a Jinja2/HTMX UI for end-to-end ingestion, querying, and evaluation.

ResearchFlow — **Multi-agent research system**

- Designed a multi-agent architecture in LangGraph orchestrating 4 specialized agents (Coordinator, Search, Reader, Writer) to research and synthesize sourced answers to complex queries.
- Constructed a FastAPI backend with live WebSocket progress updates and a SQLite persistence layer, shipping an end-to-end product from orchestration to frontend.

RESEARCH & PUBLICATIONS

LSTM Neural Networks for Stock Market Technical Analysis — *IEEE, ISMSIT 2023*

- Co-authored published research applying LSTM neural networks to stock price forecasting, benchmarking performance against Random Forest, SVM, and traditional time-series methods.

EDUCATION

Bachelor of Technology, Computer Science Engineering *July 2024*

Symbiosis International University, Pune

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Neural Networks, Software Engineering

LANGUAGES & TECHNOLOGIES

Languages: Python, SQL

GenAI / LLM: LangChain, LangGraph, LlamaIndex, RAG Architectures, Prompt Engineering, Vector Databases (Pinecone, pgvector), PyTorch

Cloud & DevOps: AWS, GCP, Docker, MLOps, Git

Full-Stack & Data: React, Node.js, FastAPI, MySQL, PostgreSQL